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Interface of Weather and Climate: The Great Midwest Flood of 2008

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ABSTRACT

Ten days of continuous torrential rain pushed warm late Spring runoff in deltaic patterns far into Lake Michigan over two subsequent months. Repeated expeditions on the *RV Neeskay* documented shallow lenses of plankton-enriched water spreading across the surface at 1 km (0.6 mi) per day for months, reaching halfway across the lake, and often running against prevailing currents. Floating cohesively near the surface, a trail of strong, river-harbor-originating biological productivity long outlasted chemical evidence of stormwater efflux in depleted coastal and offshore regions. Only sinking algal populations descended into cooler midwater depths. Previously depauperated siliceous protein- and lipid-rich diatom algae were strongly favored, suggesting a change in water chemistry not readily detected in standard water quality measurements. Initial community outgrowth in near-surface water was evident through turbidity and enhanced biomass as a wave of growth moved offshore, trailed by resurgence of midwater diatom community blooms. Anomalously high surface chloride moved past 25 km 30 days after peak flow. Conductivity is a valuable tool for detection of riverine and harbor water intrusion, as runoff-fed rivers have higher dissolved salts than receiving waters. Endurance of impact indicated that sources far from intakes could pose water quality risks for stable substances. Seasonal temperatures controlled river-lake interactions as Spring runoff warmed by surfaces was buoyant on cool lakewater. Nearshore water treatment intakes thus lay well below the initial surface lens outflows, but biological products would have impinged on them during subsequent sinking. Operational recommendations for predictive event assessment by water producers are provided.

1 | Introduction

Surface freshwater and subsurface water (groundwater) supply United States populations with potable water. Each of these is susceptible to depletion and contamination. For example, scarcely a week goes by without news regarding major flooding somewhere in the United States (Donat et al. 2016; White and Norton 2025). Wisconsin has had a sequence of

notable inland floods (Foran 2024), and many of these become Lake Michigan contributors through river runoff and direct rainfall. Besides the physical damage that intense flooding causes (NOAA 2024), its water is derived from agricultural, industrial, and urban surfaces and hence moves copious quantities of often uncommon solutes, such as trace metals; biocides of many varieties; stable organic contaminants (e.g., PolyFluorinatedAlkylSubstances, PFAS; PolyChlorinated

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Article Impact Statement

River and Coastal water intakes receive unusual, often hazardous biological particles and dissolved toxins during outwash from flood events. Physical and chemical monitoring can facilitate response.

Biphenyls, PCB; and Polycyclic Aromatic Hydrocarbons, PAH); and biological products (e.g., toxins including microcystins and ozone-consuming triglycerides, Wang et al. 2025). Solutes may directly, through intakes, or indirectly through the production of biological or geochemical products, offer challenges to water purification and distribution systems (Talbot et al. 2021). Foreknowledge and predictability are crucial for effective management (Hollister 2024). This study aims to identify measurable characteristics of source and intake water that give details and warning of anomalous ecosystem occurrences (Gründemann et al. 2022) that may contain any of these further but analytically more challenging risks.

Ten million people use potable water directly derived from Lake Michigan (Illinois EPA 2025). During the more than 10,000 days since our Southwestern Lake Michigan time series began in

1997, flood-scale Milwaukee River flows $> 140 \text{ m}^3/\text{s}$ ($5000 \text{ ft}^3/\text{s}$) occurred only 35 days, often in clusters (USGS 04087000, c.f., Westra et al. 2014). Nine of those days were during the $250 \text{ m}^3/\text{s}$ ($8828 \text{ ft}^3/\text{s}$) flood of 2008 (Midwest Climate Watch 2008) described herein as an example of consequences for receiving waters, and 12 of the remaining events were during 2018 or later, here and nationwide (NCEI/NOAA 2023). In fact, a sudden deluge in SE Wisconsin on August 10, 2025, produced a devastating 11-ft Milwaukee River level in less than 1 h at night, reaching over $450 \text{ m}^3/\text{s}$ ($16,500 \text{ cfs}$) for 9 h. These contrasting storms, one a very high flux in a short time (August 2025) compared to this protracted flow over many days, each have characteristics of value when considering intake operational preparation and/or response. How does this, here and in many similar watersheds and receiving waters, affect their ecosystem services? How can we know?

Three rivers that feed Milwaukee Harbor and dominate riverine inputs to Lake Michigan (Figure 1) drain very different landscapes with very different chemistries: one from pristine Northern forests, through farmland, into suburban and light industry, and past urban areas (Milwaukee River, MKE); one from heavy industry—tanning, Harley-Davidson, foundries, plating, paper (Menomonee River, MEN); to a shorter suburban-only source, past an international airport with seasonal deicing

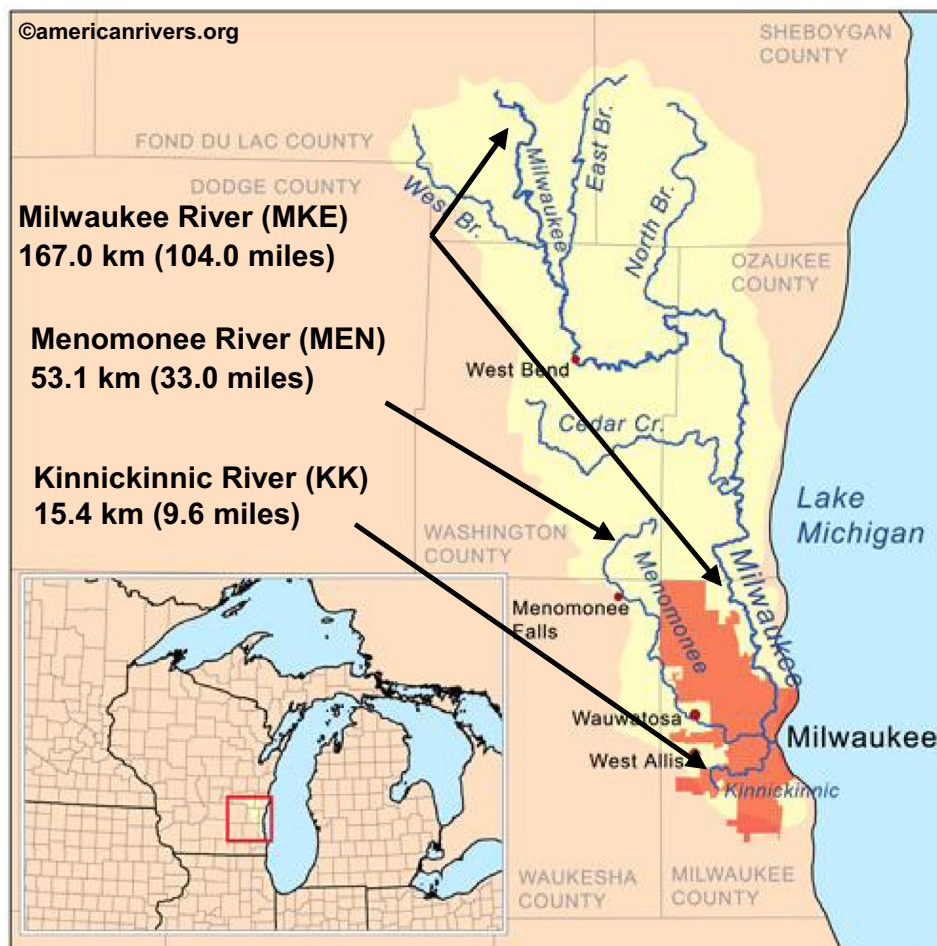


FIGURE 1 | Southeastern Wisconsin watershed showing Milwaukee, Menomonee, and Kinnickinnic Rivers that flow through the Milwaukee Area of Concern into Lake Michigan. <http://americanrivers.org/>.

into the same urban areas (Kinnickinnic River, KK). These meet and *may* mix depending on thermal structure in the Junction (JCT) of the three rivers, flow through a short linear channel, disperse into a harbor, and seep into coastal Lake Michigan (Methods Figure 3). Their legacy sediments and the modern water flowing over them carry vastly different loads of soluble materials of terrestrial origin, as mentioned above, that have fates dependent upon physical processes in the nearshore. A million yd³ (750,000 m³) Area of Concern (AOC) sediment dredging project is slated for the late 2020s (WDNR website 2024), and its deposits will reside in the Southern Outer Harbor (OHS, Figure 3). At present (2025), it is the uninvestigated release of *solubles* from that historical sediment disturbance that drives this study—that is, what processes will distribute which wanted or unwanted materials to nearshore waters containing water treatment intake sites?

The authors are dedicated to elucidating interdisciplinary factors, that is, physical structure and geology, meteorology, chemistry, optics, and biology–ecology as components of source water dynamics in large basins. Between human ecosystem services and fisheries biology, river–harbor–nearshore ecosystems depend upon the sum of these factors for their integrated productivity.

Extreme climate events range from ephemeral to permanent in their local environmental impact. Lake Michigan is a very large aquatic ecosystem spanning 4° of latitude and containing 4700 km³ (1130 mi³) of fresh water, average depth 87 m (285 ft), in a definitively seasonal midlatitude location (42°N–46°N). As such, it has distinct, reproducible seasonal patterns and substantial physical and biogeochemical inertia. Events that can leave a detectable mark on its offshore waters, beyond those routinely associated with 25°C (77°F) seasonal oscillations in surface temperature, must be sizeable. But, like the Mauna Loa pCO₂ data of Keeling (1958, 1961); Scripps CO₂ Program (2025), annually repeated seasonal cycles may be overlain on long-term consistent change, leading in this case to the concept of global warming. In Lake Michigan, persistent seasonal cycles of dissolved silicate, a diatom requirement, remained through an abrupt reduction of *magnitude* because of diatom grazing losses from Quagga Mussel feeding (Cuhel and Aguilar 2013). Basin-wide ecological transformations from previous diatom abundance to near absence were thus readily implicated by a simple colorimetric measurement of silicate (Cuhel and Aguilar 2013). Continuous intra- and inter-annual monitoring provides a necessary context from which to determine change. For Lake Michigan, both government (NOAA, EPA, USGS; c.f., Madenjian et al. 2002) and academic time series have such endurance (Cuhel and Aguilar 2013). The Cuhel-Aguilar effort is notable in that it includes sampling offshore throughout all the seasons, including deep winter, consistently for 22 of 25 years, though it is limited to a 90 km (56 mi) radius around the Milwaukee Harbor. In addition to context, it is necessary to either predict or rapidly respond to potential systematic variations.

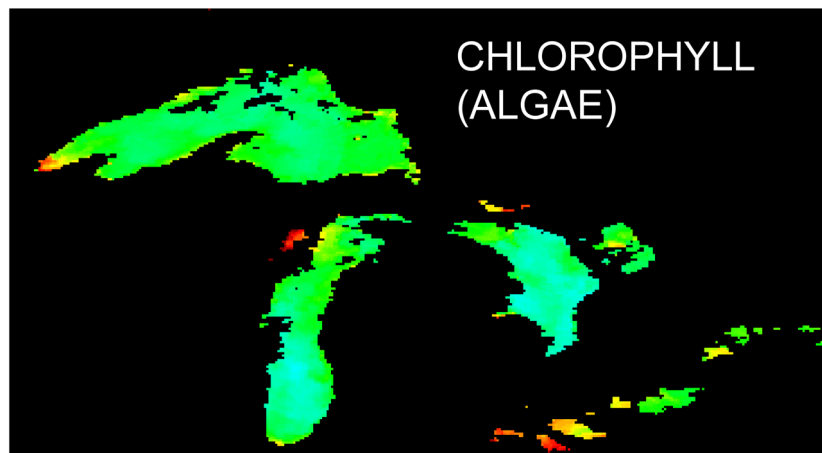
Management practices stemming from policies such as the Great Lakes Water Quality Agreement (GLWQA 1972, 2016) have also altered optical and hence heat budget properties of Lake Michigan over decade scales. Formerly episodic biological alterations (e.g., sequential invasive mussel establishments:

Fahnenstiel et al. 1995, 2010), one now persistent, also changed optical and heat transfer characteristics in addition to decimating fisheries resources. Every photon that enters water and is not outwelled ultimately becomes heat. Solar energy is absorbed by soluble and particulate components as well as by water itself (Soppa et al. 2019), and when water clarity is greatly increased by management (GLWQA nutrient limitation) and/or biology (invasive Quagga Mussel water clearing), deeper light penetration produces deeper warming (Pilla et al. 2020). Light energy penetrates well below the primary thermocline, where water intakes are situated, with eventual consequences for turbulent mixing of external sources. This was particularly well demonstrated for large lakes in winter when water temperatures are near 4°C (39°F) and small changes distinctly create or dispel physical structure (Austin et al. 2025). Extreme climate events and biological invasions may thus be synergistic (Diez et al. 2012; Ravi et al. 2022).

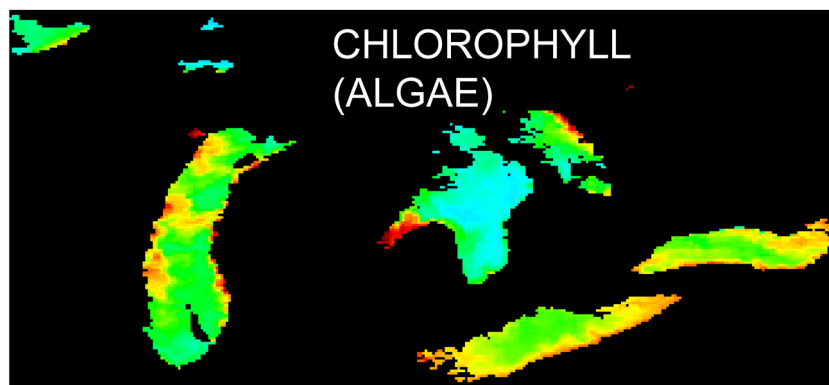
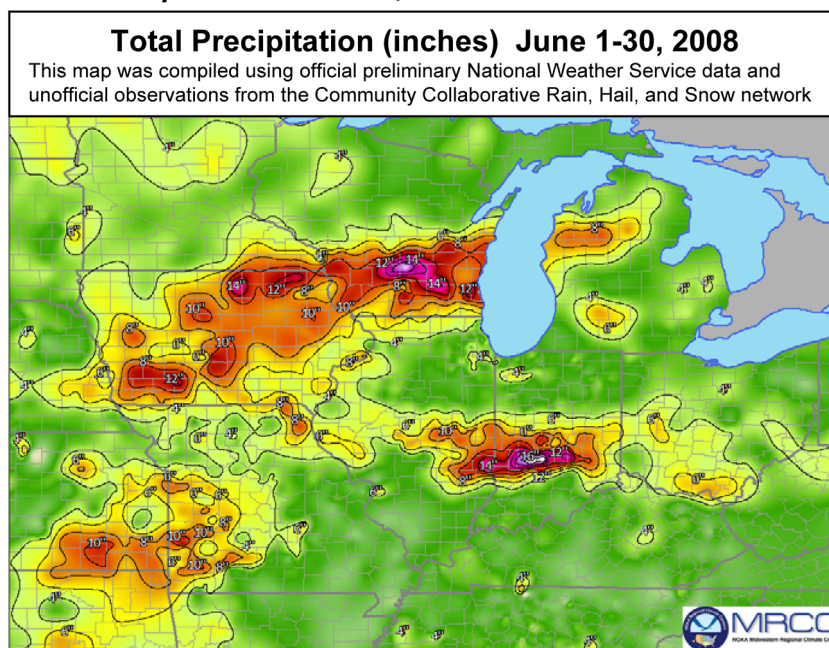
This paper focuses on the Great Midwest Flood of 2008. River floodwater injection of phytoplankton themselves, not limiting nutrients, carried organisms out onto pristine waters to bloom, as visualized in Figure 2. Satellite-derived chlorophyll (*William Balch, Bigelow*) for the five Great Lakes in early July for the year prior to the flood (2007); a contour plot of the cumulative precipitation across the region; and the same Julian date 1 month after the flood in 2008 set the stage for a major weather event responding to cumulative climate alteration in the mid-latitudes (Figure 2).

This interdisciplinary study of the Great Midwest Flood is a relatively high frequency, very detailed analysis of nearshore to midlake characteristics over time following a significant precipitation event (Table 1). It was an extreme example of an increasingly common occurrence—flooding from watersheds into municipal source waters. What those floodwaters do and do not carry is of consequence when regional potable water producers have source water intakes near effluents. Hence, *source water protection* in this context is the ability to detect and respond to ecosystem-driven changes in reactive constituents of freshwater reaching water treatment plant intakes. This is not unique—in fact, there is a substantial local history of great floods (Prein et al. 2017; Foran 2024), but few have had the level of time series study as 2008. Exemplary outcomes of the flood empirical investigations, relevant to local source water understanding and hence protection, identify methods and effects obtained during spring, summer, fall, and winter seasons following this event. The relevance of each component to synoptic visions of source water characteristic changes is described in model terms applicable to intake stations in many very large lakes.

The principal aim of this unplanned study was to link physical, hydrographic, chemical, meteorological, optical, and biological responses to a major weather event in a manner enabling predictive assessment of impacts on effluent-influenced nearshore waters harboring water treatment plant intakes. Objectives included mapping of discharge flows through adjacent coastal waters; assessment of water layering that constrains chemical and biological distributions; quantitation of algal and biomass nutrient dispersal; and timing of algal productivity changes leading to new community structure.



July 10-12 2007; "NORMAL" YEAR



July 9-11 2008; THE GREAT FLOOD OF 2008

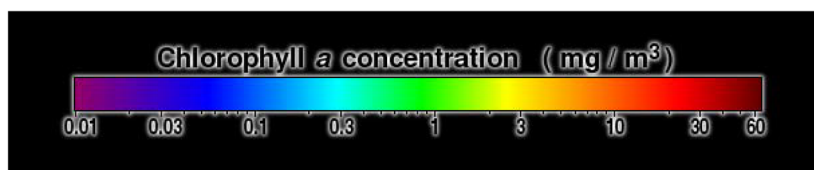


FIGURE 2 | Legend on next page.

FIGURE 2 | Laurentian Great Lakes surface chlorophyll *a* contours (mg/m³; scale at bottom) derived from satellite imaging for July 10–12, 2007 (top panel) and July 9–11, 2008 (bottom panel and scale bar) by William B. Balch, Bigelow Laboratory for Ocean Sciences. The two larger red zones on the Western Lake Michigan coast are Milwaukee to the South and Manitowoc-Two Rivers to the North, with smaller flushing from the Sheboygan River just north of Milwaukee, the Sturgeon Bay Canal, and Death's Door at the Northern end of Green Bay (southern part obscured by clouds, black). Center Panel: Cumulative Midwest precipitation contours during June 01–15, 2008 in cumulative inches (red = 18, green = 5) from NOAA Midwest Regional Climate Center weather data. Storm crossed Lake Michigan into central Michigan state.

2 | Materials and Methods

2.1 | Setting

The lake originally formed due to the separation of two tectonic plates along the superior Craton, producing a mid-continent rift. The Laurentide Ice Sheet, a massive glacier, covered much of the area over a Paleozoic shallow carbonate-bedrock sea and other areas known today as the Great Lakes (EPA 1995). About 14,000 years ago, the glacier that occupied Lake Michigan began to pull back from a large arcuate highland that flanked the southern part of the lake basin (Chrastowski and Thompson 1994). The Mid-Lake Reef Complex (MLRC), a large area of erosion-resistant staircase limestone reefs (Paleozoic coral relics 40–60 m rising steeply out of 150 m deep surrounding basins, Figure 3), is littered with ice-transported gravel, cobble, boulders, and sand. The formation behaves as a seamount and affects basin-scale circulation dynamics. Limited existing biological and physical evidence suggests that seamount-like upwelling, or Taylor columns, lead to active water transport (Figure 7 in Cuhel and Aguilar 2013; Houghton et al. 2010; Possamai et al. 2024).

Invasive species marked substantial changes in the ecology of Lake Michigan. Sea lamprey (*Petromyzon marinus*), initially parasitic on Lake Trout, invaded Lake Michigan from the Atlantic Ocean through shipping canals in 1936, devastating Great Lakes fisheries (Adams et al. 2021). Other invasive species with decimating impacts include the 2003 establishment of the quagga mussel (*Dreissena rostriformis bugensis*), which caused the rapid collapse of most commercial fisheries in Lake Michigan (Cuhel and Aguilar 2013; Rowe et al. 2015), in part because of their use of MLRC circulation for basin-scale grazing success. Invasions and resultant ecosystem modification affected regional and lakewide water characteristics.

2.2 | Environmental Sampling

Lake Michigan samples were collected on expeditions of the 71-ft *Research Vessel Neeskey* between March and October of 2008 as part of a then 10-year continuing time series. Additional expeditions, particularly to the offshore Mid-Lake Reef Complex (MLRC, Figure 3), were made with undergraduate research programs (National Science Foundation Research Experience for Undergraduates [REU] and NSF Undergraduate Biology-Mathematics Mentoring [UBM]). A Wisconsin Sea Grant funded study to the late Dr. Craig Sandgren and Dr. John Berges collected additional Fox Point (FP) reference station electronic data acquisition instrument profiles (CTD, Conductivity-Temperature-Depth, detailed below), surface, and 25 m samples

for the authors during the summer. The area of study with cruise tracks for MLRC visits used in this study and locations of the Fox Point 100 m deep reference station, city water intakes Linnwood Water Treatment Plant (LW) and Howard Avenue Water Treatment Plant (HA), and Milwaukee Area Of Concern (AOC) river-harbor stations are detailed in Figure 3. Stations are designated by bottom depth on the transect line (e.g., LW20 is the 20 m deep station off the Linnwood Water Treatment Plant and near the location of the intake).

Transect samples were collected from a deck box receiving 2 m water from below the ship's hull using Teflon-lined stainless steel tubing with an impeller pump. Transit from inlet to collection took 6 s, which was 25 m at 9 knots (17 km/h, about one ship-length). Hydrographic information on transects was collected from the deck box with a Hydrolab 5A datasonde (Hydrolab Inc., TX), yielding temperature (*T*, °C), specific conductivity (SpCond, μ Siemens/cm), pH, in vivo algal fluorescence (Volts), dissolved oxygen (DO₂, mg/L), oxidation-reduction potential (ORP, mV), and turbidity (Nephelometric Turbidity Units, NTU); data were recorded at 1 s intervals synchronized to the ship's navigation system for precise location assessment. Station hydrographic profiles were determined at higher sensor frequency (to 8/s) and precision (to 4 decimals) with a Seabird 25 Sealogger CTD (Conductivity-Temperature-Depth, SeaBird Electronics, WA) providing depth (m), temperature (*T*, °C), specific conductivity (SpCond, μ Siemens/cm), dissolved oxygen (DO₂, mg/L), 0.25 m beam transmission (%*T*), in vivo algal fluorescence (Volts), photosynthetically active radiation (PAR, μ Einsteins/m²/s), oxidation-reduction potential (ORP, mV), and pH. Data were downloaded upon retrieval, processed using the manufacturer's graphical software, and inspected for suggested sampling additions. Later, the manufacturer's detailed data conversion software was used to produce high precision outputs that were bin-averaged to 0.1 m (at < 20 m stations) or 0.25 m resolution as used here.

A historical depth series updated in 1997 (0, 5, 10, 15, 20, 25, 30, 35, 40, 45, 60, and 90 m plus a sample collected 1 m above the sediment-water interface (SWI, B-1) using an upside-down Niskin bottle attached to the CTD frame with a peg-leg trip pole) was always collected as station depth and weather allowed, with additional depths taken based on CTD profile information. CTD light was occluded to 20 m when lowered on the shade side of the vessel. Manual Photosynthetically-Active Radiation (PAR) attenuation was additionally assessed on the sun side of the vessel at 0.5–1 m intervals to 27 m using an underwater 4 π Quantum sensor normalized to incident flux from a cosine sensor mounted on the railing (LI-COR Inc., NE).

On-station water column samples were collected by bucket (surface, 0 m) and with 5-L Niskin bottles (General Oceanics, FL)

TABLE 1 | Dates, locations, and number of samples collected during 2008 flood study on Lake Michigan.

Date of trip	FP	LW20	LW30	GC06	GC10	GC20	GC30	OHN	OHS	JCT	NE	SR
October 15, 2008	—	—	—	—	—	GC20	—	—	—	—	—	—
October 13, 2008	FP	LW20	LW30	—	—	—	—	OHN	OHS	JCT	—	—
September 25, 2008	FP	LW20	—	—	—	—	—	OHN	—	JCT	—	—
September 9, 2008	FP	(LW20)	—	—	—	—	—	—	—	—	—	—
September 2, 2008	FP	—	—	—	—	—	—	—	—	—	NE	SR
August 27, 2008	FP	(LW20)	—	—	—	—	—	—	—	—	—	—
August 11, 2008	FP	LW20	LW30	—	—	—	—	OHN	OHS	JCT	—	—
August 7, 2008	FP	(LW20)	—	—	—	—	—	—	—	—	—	—
August 4, 2008	—	—	—	—	—	GC20	GC30	—	OHS	—	—	—
July 24, 2008	FP	(LW20)	—	—	—	—	—	—	—	—	—	—
Jul 21, 2008	FP	—	—	—	—	—	—	—	—	—	—	SR
July 10, 2008	FP	—	—	—	—	—	—	—	—	—	NE	—
July 7, 2008	FP	(LW20)	—	—	—	—	—	—	—	—	—	—
July 2, 2008	—	—	—	—	—	—	—	OHN	—	JCT	—	—
June 30, 2008	—	—	—	—	—	GC20	—	—	OHS	JCT	—	—
June 25, 2008	FP	(LW20)	—	—	—	—	—	—	—	—	—	—
June 12, 2008	—	—	—	—	—	GC20	GC30	OHN	OHS	JCT	—	—
June 10, 2008	FP	LW20	LW30	—	—	—	—	OHN	OHS	JCT	—	—
May 22, 2008	—	—	—	—	—	GC20	—	—	OHS	—	—	—
May 20, 2008	—	LW20	—	—	—	—	—	OHN	—	—	—	—
May 14, 2008	FP	—	—	—	—	—	—	—	—	—	—	—
April 30, 2008	FP	LW20	—	—	—	—	—	OHN	—	JCT	—	—
April 8, 2008	—	—	—	GC06	GC10	GC20	GC30	OHN	OHS	JCT	—	—
March 26, 2008	FP	LW20	LW30	—	—	—	—	OHN	OHS	JCT	—	—

Summary of collected samples in 2008

	FP	FP	LW	GC	MLRC
	US	SAND			
Number of visits	10	6	6	6	3
Likely sample depths	150	18	24	24	153
Likely different analytes	2550	306	408	408	2601
CTD casts	80	—	—	—	17
Par profiles	122	—	—	—	22
PVSI samples	158	—	—	—	62

Note: Locations: FP, Fox Point; LW, Linnwood; GC, Green Can; OH, Outer Harbor; JCT, Junction of rivers; NE, Northeast Reef; SR, Sheboygan Reef; MLRC, all offshore reef stations; CS, Dr. C. Sandgren samples; CTD, Conductivity-Temperature-Depth instrument; PAR, Photosynthetically-Active Radiation; PvsI, Photosynthesis rate measurements. (L20) are CS samples with CTD, surface, and 25 m water only.

The first flood sampling, during the last day of rainfall, was June 10, 2008.

for profiling work, while transect samples were collected from the outflow of the deck box. Comparison of deck box and 2 m Niskin samples for chemistry and biological rate processes (e.g.,

photosynthesis) was excellent over the years. A summary of all dates, stations, and approximate numbers of samples and analytes used in this flood year presentation is shown in Table 1.

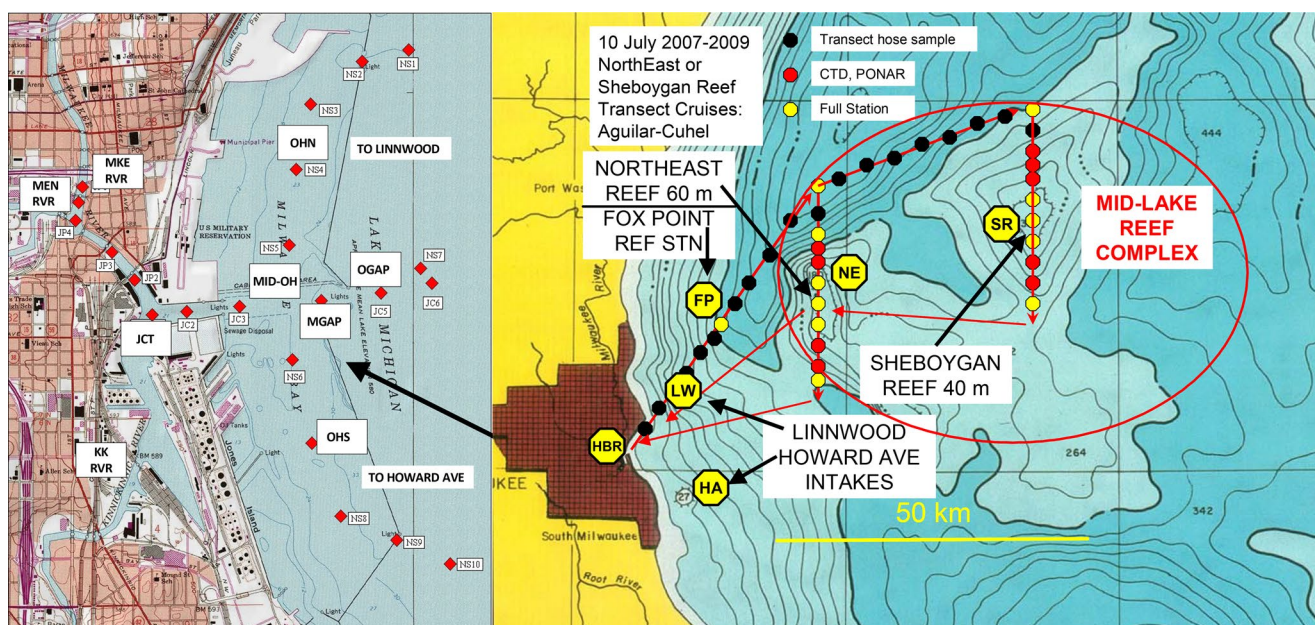


FIGURE 3 | Map of Southwestern Lake Michigan and Milwaukee Harbor with sampling transects and station types. Left Panel: Milwaukee Harbor Area of Concern, with river stations (Figure 1): Confluence (JCT), mid-harbor channel (MID-OH), Outer Harbors North and South (OHN, OHS), and Main and Outside Gaps (MGAP, OGAP). Includes regularly visited stations not included in this study. Right Panel: Intakes Linnwood (LW) and Howard Avenue (HA), Fox Point time series reference station (FP); Northeast Reef rise and plateau (NE); Sheboygan Reef rise and plateau (SR); CTD, conductivity-temperature-depth instrument lowering; PONAR, bottom grab; black circles are in-transit 2m samples. Contours in fathoms with lighter = shallower. Depth numbers in feet. Contour map by Ratko Ristic.

2.3 | Laboratory Analysis

Whole water for particulates, microscopy, photosynthesis, and other analyses was stored on the vessel in rinsed 4-L polycarbonate (PC) bottles. Subsamples were immediately syringe filter-sterilized (0.2 μ m pore size Supor-200 polyethersulfone; Pall Inc., MI) into 50mL conical tubes and other containers for labile analytes. All water was refrigerated or iced for transit. Chemical analyses were accomplished by manual test tube or flow injection (FIA) methods according to APHA (2002). Particulate P [PP] (Supor 0.2 μ m), chlorophyll *a* [CHL*a*] (GF/F <0.7 μ m), and particulate silicate (0.8 μ m PC) were analyzed by persulfate-molybdate, DMSO:90% Acetone fluorometry (Shoaf and Lium 1976), and a modification of the carbonate extraction, silico-molybdate assay of Paasche (1980) respectively.

Perkin-Elmer Lambda 25 dual-beam and Beckman D7000 diode-array spectrophotometers using one or two of 1, 2, 5, and 10 cm cells (depending on the required range of analyte concentrations) were used for colorimetric methods. Standard curves of eight or more concentrations were extended beyond linear response for each assay to quantify unexpectedly high samples. Multiple curves of standards were individually read before, among, and after total run samples to account for rare drift, with bracketing curves averaged and regressed for quantitative analysis.

Dissolved Inorganic Carbon (DIC) was analyzed by the Teflon FIA of Hall and Aller (1992). Ammonium was measured by phenol-hypochlorite (Koroleff 1976). All methods were run in duplicate or better with inclusion of certified standards (CERT) for accuracy assessment; rarely, assays yielding >5% error in accuracy were inspected and repeated if possible. For most

analytes, the error of duplicates and accuracy (CERT recovery) were reliably 2% or better. Excel statistics included averaging of standard curves and linear regression—regression quality $r^2 < 0.999$ was inspected for aberrant standard points and/or non-linearity. Duplicate or greater replicates of sample analyses were calculated independently, averaged, and error calculated from means. Inspection of errors occasionally revealed keyboard input errors when replicates agreed poorly (>2%–5%). Graphical visualization includes error bars as determined above, but at 2% precision and the range of values across sample venues, error bars were inside a #7 Excel symbol. Contour plots and specialty statistics were derived with Systat programs.

2.4 | Photosynthetic CO₂ Fixation

Photosynthesis was measured during the daytime of the next day after sampling unless sufficient daylight was available immediately after return. This avoided interfering with circadian rhythm behavior in the physiology of especially diatoms and dinoflagellates (Cuhel et al. 1984; Häfker et al. 2023). The Photosynthesis-Irradiance (PvsI) approach of Lewis and Smith (1983) was 1–3 h temperature-controlled ¹⁴C-bicarbonate (MP Biomedicals, OH) light gradient incorporation with 12 light intensities, two darks, and two zero time controls per sample (c.f., Fogel et al. 1999). Measured input variables included extracted CHL*a* profiles, in situ PAR extinction, in situ temperature profile, DIC, ¹⁴C activity, and experimental PAR for up to 12 depths using up to three temperatures (in situ $\pm 2^\circ\text{C}$). Calculations used the programs of Fee (1990): in sequence (1) PSPARMS to derive initial slope of photosynthetic efficiency and maximum potential rate of photosynthesis; (2) DPHOTO to use known daily latitude-dependent PAR

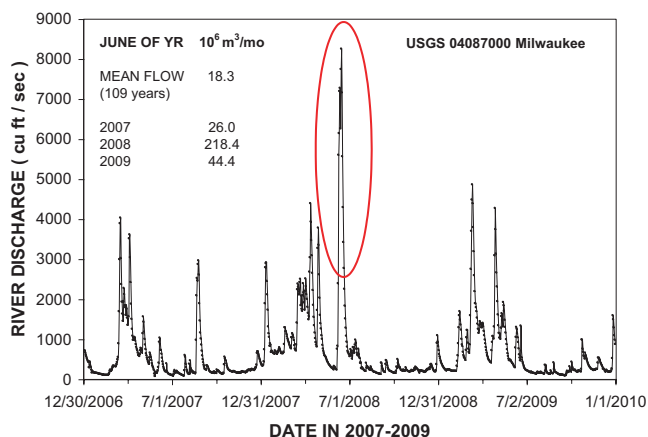


FIGURE 4 | Milwaukee River flow at Estabrook Park USGS Gauging Station 04087000 in ft³/s (USGS Units) for 2007–2009 with inset table of 10⁶ total cubic meters for the month of June of each year and 109 year average. Red oval identifies 2008 flood outcome.

(30 min intervals), measured PAR extinction with depth, and PSPARMS output to calculate daily photosynthesis profiles; and (3) YPHOTO-YTOTAL to calculate daily areal (per square meter) photosynthesis over a year with consideration of seasonality and stratification effects on productivity. The data span light intensities from 5 to 1000 μ Einsteins/m²/s and are suitable for modeling.

3 | Results and Discussion

3.1 | Weather Setup

More than a half-meter (1.6 ft) of rain fell on central and southern Wisconsin during a slow-moving storm in the first 2 weeks of June 2008 (Figure 2 above). The actual Lake Michigan watershed in Southeastern Wisconsin near Milwaukee is narrow (EPA 1995), so the cumulative precipitation to the local Menomonee and Kinnickinnic Rivers was less than inland, but the south-flowing Milwaukee River drains substantial northern lands and carries their waters to the Milwaukee Harbor (Figure 1). The rain event that started on June 01, 2008 was detected as increased flow on the Milwaukee River on June 05, a 4-day lag, and increased to a maximum of 234 m³/s (8260 ft³/s) on June 14, over a 100-year average of 7 m³/s (247 ft³/s) (Figure 4) at the lower Milwaukee River (USGS Gauging Station 04087000). More than 140 m³/s (5000 ft³/s) persisted for 9 days in a row, and June yielded a flow of 218 × 10⁶ m³ (7.7 × 10⁹ ft³) for the month, over 10 times the long-term average. This places the storm as an extreme meteorological event.

One month earlier, on 30 April 2008, the Fox Point time series station (FP) was still mixed to the bottom (3.3°C [37.9°F]; Table 2). By 02 June 2008, surface water had warmed to just above 5°C (41°F) and the first weak signs of structure at 10 m (32.8 ft) had set in, but there were no inflections in transmission and the light field was uniform at 1% I_0 (surface-penetrating PAR) = 43.6 m (143 ft). (Data not shown, Sandgren-Berges visit).

4 | Initial Gradients During Outwash

4.1 | Onshore-Offshore Surface Constituents

Our expedition on June 10 was during the last day of actual precipitation but prior to maximum flow, and a transect out to the 100 m (328 ft) FP station demonstrated a discernable harbor influence in near-surface water chemistry and optics over the LW water intake site. Pre-, during-, and post-maximum flow offshore transects for some biogeochemically relevant parameters are discussed below (Table 2). Sampling dates and details are found in Methods (Table 1).

During the last days of the two-week downpour (June 01–15, 2008), we provided national Research Experience for Undergraduates students with an expedition from the confluence of three Southeast Wisconsin rivers (JCT) through the Outer Harbors North and South, past the Linnwood Water Treatment Plant Intake (LW20; 7 km northeast of the Main Gap) and out from the slightly more offshore 30 m contour (LW30; 8 km) to our long-term monitoring station FP at 25 km (Table 2 and Figure 3). Linnwood and Fox Point are both upstream of the prevailing current moving past Milwaukee from north to south (Beletsky and Schwab 2001). Distinct evidence of rain runoff, differing from groundwater inputs, was evident in river and outer harbor station surface (2 m) water from the ship's Teflon pumping system. Notably, a second city intake exists 4.5 km south of the harbor (Howard Avenue, HA) and is in the direct path of harbor water flowing south with the often sub-surface along-shore current. In this paper, our interest is primarily directed towards the upstream LW city water intake because of the anomalies associated with harbor water getting there at all, and complementary to the downstream HA intake.

Chemicals abundant in rain itself and direct runoff in this region were observed to follow surface spatial gradients from river sources into Lake Michigan for the first expedition (Table 2). Ammonium (NH₄⁺) and nitrate plus nitrite (NO₃⁻ + NO₂⁻), both abundant in spring rain and agricultural runoff, were typically high in river and harbor water but decreased rapidly to offshore lake levels. Orthophosphate (SRP) is more sourced from agricultural fertilization but was also rapidly depleted in nearshore waters. Total Phosphorus was enhanced over open lake levels as far as 6 km offshore during this early flood stage. Of particular interest was algal biomass as extracted CHLa, which was small in the flooded rivers and decreased even further into coastal Lake Michigan during runoff. During coastal warming (thermal bar in Spring, Consi et al. 2008), nearshore waters mixed to the bottom (<15 m) where they became accessible to particle scouring by newly (since 2003) dense Quagga Mussel benthic bivalve filter feeders (reviewed in Cuhel and Aguilar 2013). At this time of yearly progressions, the water column rapidly cools moving offshore and a warm intrusion can become a thin cohesive lens on the surface, immune from benthic interaction.

Peak storm flow on 5 June (2.5 × 10⁶ m³/day [288 × 10⁶ ft³/day]) was evidenced on the first expedition to the Fox Point time series station 5 days later (June 10, 2008), when chemical and biomass

TABLE 2 | Physical–chemical–biological parameters of Lake Michigan surface water for onshore-offshore gradients from Milwaukee Area of Concern rivers, through Outer Harbors, past the LW20 source water intake for Milwaukee Water Works, and offshore to the Fox Point long term time series station.

STN	YYYY/ MM/DD	MGAP (km NE)	SST (C)	SECCHI DEPTHm	Cl– (μM)	CHLa (μg/L)	NH4+ (μM)	NO ₃ +NO ₂ (μM)	dSiO ₂ (μM)	SRP (μM)	TP (μM)	DIC (mM)
Spring mixed period 30 April 2008												
JCT	2008/04/30	–1.70	11.8	1.30	1992	3.77	8.8	78.6	93.5	1.341	2.66	3.87
OHN	2008/04/30	–1.30	10.6	1.20	1579	3.79	5.0	79.1	82.7	1.204	2.36	3.48
LW20	2008/04/30	4.30	6.1	5.00	437	1.00	0.7	33.1	27.0	0.028	0.21	2.28
FP	2008/04/30	25.60	3.3	14.20	325	1.19	0.1	27.6	30.3	0.012	0.10	2.14
During last 2 days of deluge 6 days before peak river flow 10 June 2008												
JCT	2008/06/10	–1.70	20.3	0.45	1017	2.76	6.6	106.8	125.7	1.771	6.16	2.88
OHN	2008/06/10	–1.30	18.4	0.60	1050	1.51	6.5	98.6	99.2	1.635	3.46	2.82
OHS	2008/06/10	–1.20	18.1	0.80	1278	1.48	6.9	108.4	95.1	1.609	3.32	2.91
LW20	2008/06/10	4.30	12.9	1.75	503	0.43	1.5	38.1	31.3	0.033	0.66	2.11
LW30	2008/06/10	6.00	12.4	5.10	359	1.24	0.0	27.2	22.0	0.024	0.23	2.15
FP	2008/06/10	25.60	7.6	15.85	325	0.50	0.1	27.3	29.3	0.017	0.09	2.07
Two months after last heavy rain 11 August 2008												
JCT	2008/08/11	–1.70	27.4	1.25	1899	8.97	8.2	81.0	129.1	1.825	2.93	4.42
OHN	2008/08/11	–1.30	22.8	1.95	561	17.31	2.0	32.3	24.9	0.154	0.94	2.40
OHS	2008/08/11	–1.20	22.2	2.00	874	32.08	0.5	40.6	35.1	0.134	1.62	2.62
LW20	2008/08/11	4.30	21.6	11.10	330	0.79	0.9	16.7	20.5	0.004	0.14	2.17
LW30	2008/08/11	6.00	21.9	13.09	326	0.62	0.4	19.1	20.5	0.004	0.08	2.19
FP	2008/08/11	25.60	22.0	17.45	344	0.46	0.4	22.6	17.4	0.004	0.07	2.08

Note: 20, 30 are intake for Milwaukee Water Works, and offshore to the Fox Point long term time series station. 20, 30 are station depths (m) along the transect, and km NE MGAP is the distance NorthEast from the breakwall. Listed from rivers moving through harbors into offshore Lake Michigan for each date (Figure 3). Given precision is correct for each analyte.

Abbreviations: DIC, total dissolved CO₂; dSiO₂, dissolved silicate; TP, total phosphorus; SRP, soluble reactive phosphorus; SST, sea surface temperature.

gradients (Table 2) existed between Milwaukee Harbor (negative distance inside the main gap) and offshore stations. River terminus (Junction; JCT) and outer harbor (OH) surface samples were often less enriched in mineral-derived chemicals (DIC, NH₄⁺, NO₃[–], SiO₂, SRP) and algal biomass (CHLa) than prior to the storm because of storm-flow dilution and shorter harbor residence time. Solutes more important to potable water producers may equally well be in there—just because the authors did not measure that does not mean it was not there. Chloride remained at typical river level near 1000 μM. The magnitude of outflow was also evident in very shallow transparency (Secchi depth <2m up to 5km from the main gap). In late mixing and early stratification (surface temperature <6°C), the Secchi depth has been >15m since Quagga Mussel establishment, so a 1.75m observation several km offshore was shallow, indicating reflectance turbidity (see also light profiles in Figure 7 below). Elevated concentrations of conservative river indicator chloride remained in coastal waters over 4km East-North-East of the harbor mouth, even though it was only three times lakewater concentration in the river source (Junction). For nutrients, uptake may be distinguished from dilution using the ratio of the

subject compound to chloride. From this perspective, ammonia and phosphate were rapidly assimilated while silicate and nitrate were not. This is consistent with long-term observations of phosphate in the very limiting 20 ± 10 nM range (Brooks and Edgington 1994) while nitrate and silicate rarely drop below 20 μM.

4.2 | Harbor Water Lens Below Surface: Conductivity Evidence

In water, conductivity is electrical transmission proportional to ionic content, and is a very powerful tool for assessing sources and duration of anomalous events with real-time, high precision but low effort monitoring and/or surveys (Figure 5). For freshwater venues, conductivity is increased by flow through rocks and soils (groundwater) and seasonally in mid-latitudes by road salt runoff. In Lake Michigan mid-lake surface conductivity is about 290 μSiemens/cm. Inflow concentration from Southeastern Wisconsin rivers is often three to four times higher, driven primarily by the carbonate

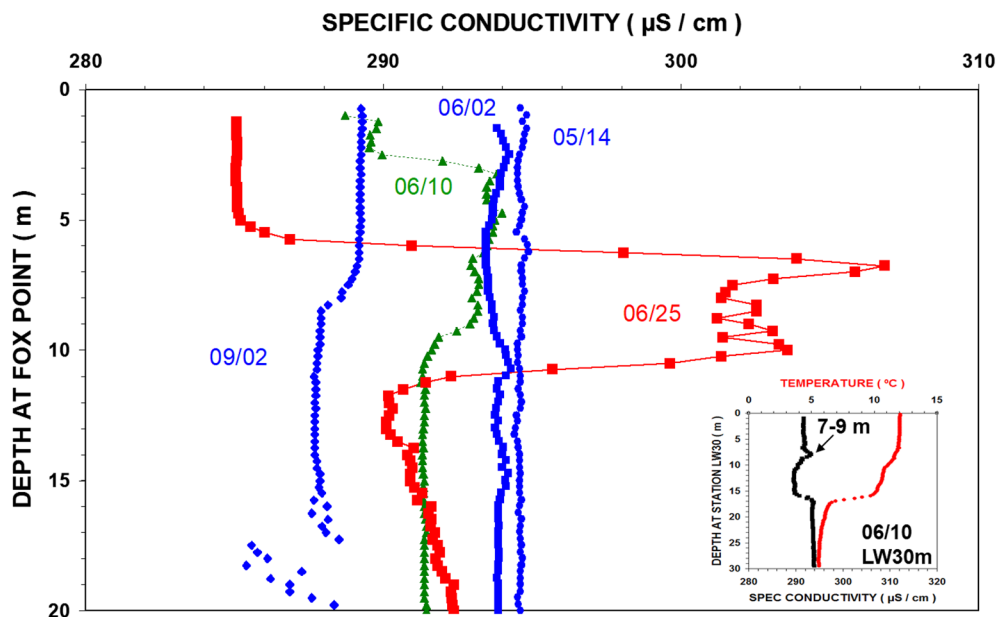


FIGURE 5 | Temperature-normalized conductivity (specific conductivity) from CTD instrument lowerings at the 100m Fox Point station, top 20m shown. Data are 8/s datasets bin-averaged to 0.25m intervals (4–10 samples/bin); precision approaches 0.1%. Dates prior to (●, ■), during (▲, ■) and following (◆) flooding are shown next to the data. Dependent axis scale is greatly expanded. Inset is of cast data from June 10, 2008 at Station LW30 inshore of the Fox Point station, showing incipient subsurface tongue of higher conductivity water prior to major changes 2 weeks later.

equilibrium of limestone weathering and seasonal salting, and therefore provides a fine signal at the harbor mouths. In the June 10 sampling, the deluge may have washed road salt off inland surfaces, though the last snow was April 12. The Linnwood Water Treatment Plant 7km north of Milwaukee Harbor occasionally experiences day-scale or longer excursions of increased conductivity at their ~20m deep intake, and there is no known source of ion-enriched water within Lake Michigan in this region. An inference then is that river-harbor effluents are moving against the prevailing southerly current and reaching the intake either directly or through sporadic upwelling events from time to time.

The Great Midwest Flood of 2008 provided an opportunity to demonstrate onshore-offshore transport over time (Figure 5). Of the 16 visits made to the 100m Fox Point time series station in 2008, several were selected to represent pre-storm winter-spring mixing (May–early June), during (June 10) and shortly following major outflows from the rivers (June 25), and several months (August and early September) after the event. Conductivity is typically higher in rivers than in large lake receiving waters, as suggested by surface chloride in the river-harbor stations ($>1000\mu\text{M}$) decreasing to open lake levels (ca. $325\mu\text{M}$) moving offshore in Table 2. In spring, river-harbor waters are likewise warmer and less dense than offshore, and this upper water temperature surplus decreases greatly in summer.

Main outflows exemplified by June 10, 2008 occurred when nearshore lake stations were 6°C less than the river-harbor source water and open lake station FP was 10°C (50°F) lower. Despite the large discharge, chloride was three times higher in outflows ($1000\text{--}1250\mu\text{M}$) than offshore FP (Table 2). Occurrence of a tongue of higher conductivity water was spatially first

evident in a CTD profile at LW30 just offshore of the 20m LW intake (Figure 5 inset): at 7–9m there was a distinct $1\mu\text{Siemen/cm}$ excursion. This was shallower than the ~20m intake and likely not noted at the plant. On the next expedition (June 25), a much more pronounced layer was found at the offshore FP station, descending to residence at 6–10m depth 2 weeks later and very evident in Figure 5. Pre-flood isohaline conditions in May and early June were followed by apparent surface water dilution (0–3m depth, Figure 5) from direct precipitation (effectively half a meter on every square meter of surface, Figure 2). This station, 25km Northeast of the Main Gap, was not close enough to display floodwater influence during the last days of rainfall, and the more inshore station LW20 was more deeply mixed at lower temperature. However, by 2 weeks later, an even lower surface conductivity was underlain by enrichment of about $10\mu\text{Siemens/cm}$ that persisted through the end of July. By September, cooling had begun to mix the water column. This sequence demonstrates that conservative solutes can move 25km or more while retaining some cohesiveness, even after 15 days of transit.

Conductivity provides a simple, inexpensive, and real-time analysis that can warn of impending change. Working with our regional water producers, there have been frequent anomalous conductivity spikes lasting days or longer at the intakes and with few surface clues. Recent year-round profiling has shown density layering even within Milwaukee Harbor (Figure 6), and this is without doubt widespread among harbors. In cases where cooler high conductivity river water (e.g., Fall) lies at the bottom (Figure 6), proper wind conditions or pressure oscillations from seiche can drive bottom water out into the nearshore. In winter, warmer river water can likewise pool at the bottom and become subject to transport. Coastal gyres and current reversals common to non-linear coastlines

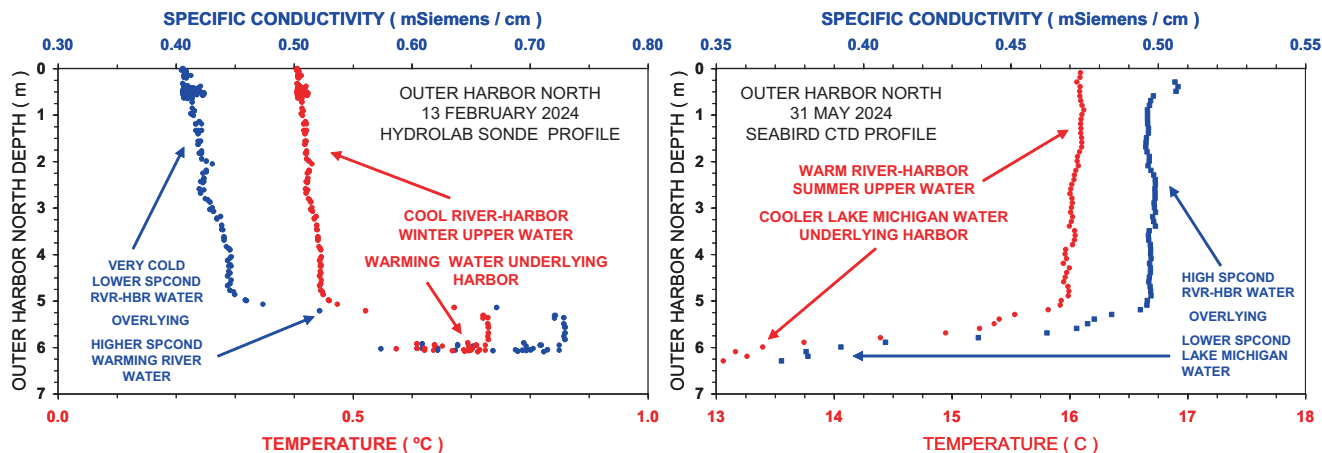


FIGURE 6 | Meter-thick bottom pools of water separated from the surface by stable density gradients in the Outer Harbor North, closest to the Linnwood intake. Left panel: Hydrolab DS5 datasonde profile in February 2024 showing winter reverse stratification at 5 m with higher specific conductivity river water at the bottom. Right panel: Summer CTD profile in late May 2024 showing cooler, lower specific conductivity Lake Michigan water below 5 m under river water.

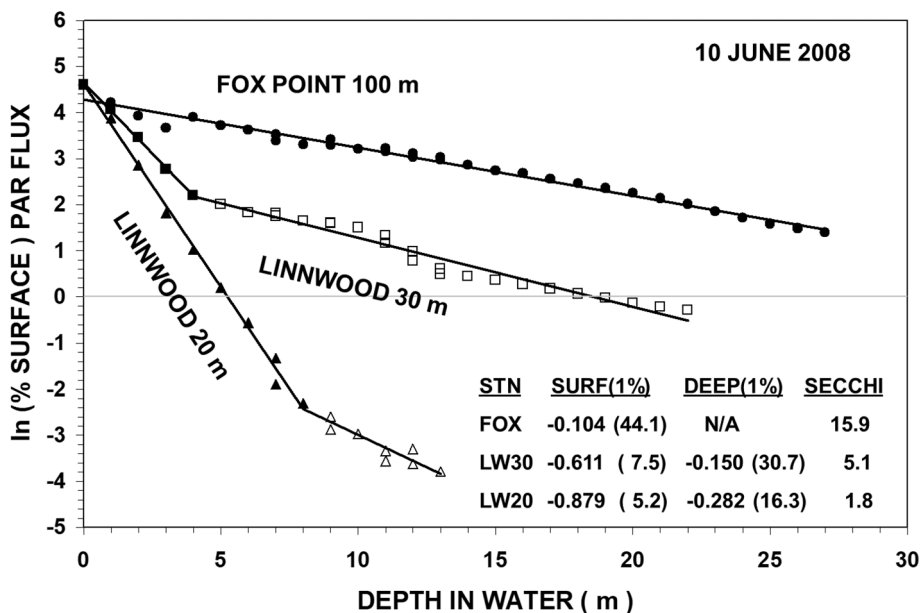


FIGURE 7 | Semi-logarithmic plot of often biphasic visible light (PAR, % surface influx) attenuation on June 10, 2008 for stations offshore of the Linnwood Water Treatment Plant. LW20, 30 are water depths at station. Steeper slopes are less transmissive water. Inset gives slope and distance of 1% penetration for upper and deeper water masses when present along with Secchi Disk visibility (m) at each station. Gray line depicts 1% level of surface inwelling radiance.

can thus drive harbor effluents along-shore towards local water intakes.

4.3 | Light Penetration and Turbidity Gradients

Biphasic Photosynthetically Active Radiation (PAR, visible light) penetration obtained with a submersible quantum meter showed 2–3 times higher attenuation in the top 4 m of the inshore water column with clearing below, where quagga mussel filtration activity was pronounced (Figure 7). CTD beam transmission (clarity) was also minimum in the upper water at LW20, increasing from 41% to 83% in a sharp step at 8 m. The upper water was

313 μ Siemens/cm (6% higher than underlying 294 μ Siemens/cm water), higher in conductivity as well, indicating a cohesive lens of turbid harbor water at the most nearshore city water intake station. The same features, more muted but still distinct, were repeated at LW30 but were not evident at the Fox Point station 25 km offshore. Together, these observations are consistent with a surficial lens of turbid harbor washout moving offshore. It is clear from Figure 7 that Secchi Depth (e.g., Table 2) is a poor proxy for light penetration when biphasic turbidity profiles are in force but is sensitive to overlying lens conditions.

Visible light gradients (PAR) are influenced by different types of particles and dissolved materials, and can be of great value

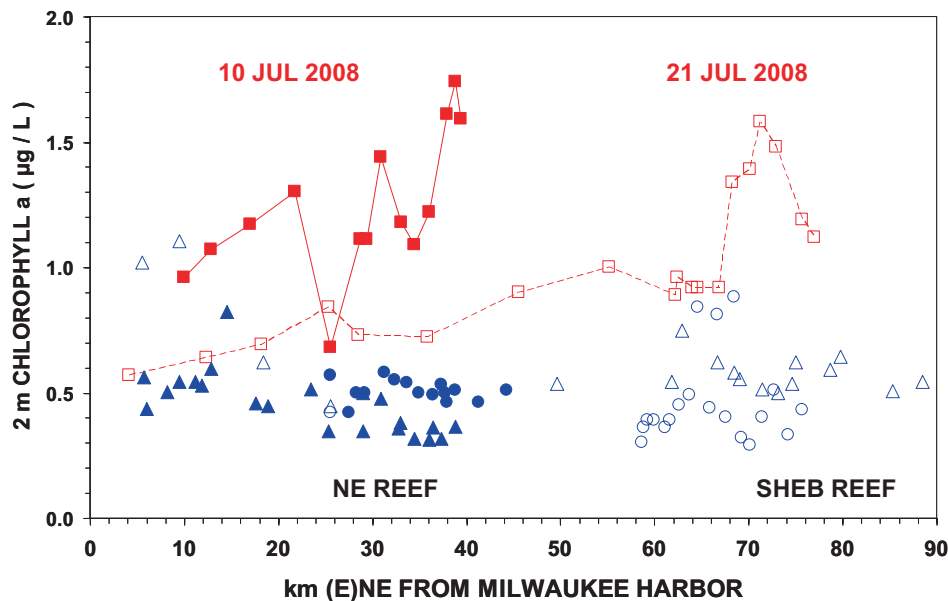


FIGURE 8 | Transit box (2m) extracted chlorophyll *a* during annual transects from Milwaukee Harbor (0km) to the NE Reef (40km; solid) on 10 July and to Sheboygan Reef (75 km; open) on 21 July 2007 (circles), 2008 (squares), and 2009 (triangles). Down-reef (southerly) samples are lower distances (Figure 3) on each transect.

in source water protection by identifying inconsequential vs. hazardous turbidity events. Clays and sand are often suspended in nearshore waters, whether by runoff, storm suspension in shallow waters, and/or human involvement. They are predominantly opaque and reflective, and of greater consequence for *turbidity* than for *transmission* (see recommendations section). Phytoplankton are dark and usually translucent but absorptive. Colored Dissolved Organic Matter (CDOM) is only absorptive and at compound-specific wavelengths. Particles are often layered in water columns while CDOM may be more uniformly distributed.

5 | Lakewide Evidence for Cohesive Outflow

In the 2008 satellite chlorophyll image (Figure 2), a visually apparent wide areal distribution of nearshore river effluent indicated increased and cohesive flows throughout both the western and eastern coasts of Lake Michigan as the storm moved over the region. Virtually every river mouth is the site of a town or city that uses source water from the receiving body, here Lake Michigan. Chicago IL area rivers were reversed so outflows go to the Mississippi River instead of Lake Michigan. Opposing river flows from Frankfort, MI, and Two Rivers, WI, merged to form a contiguous chlorophyll band across Lake Michigan (Figure 2). This is a *climate* outcome because it is an example of increasingly frequent mid-latitude strong storms.

On this scale, each of these river mouths may be 2–3 pixels at most, but plumes of outflow chlorophyll appear more like river deltas even after a month of flow. Because they are symmetrical, they must in some cases flow against or over prevailing currents. This has ramifications for near-river, especially urban water treatment plant intakes if the outflow flows over or drops below the surface. Here in June–July, river water is much

warmer (18°C–20°C) than nearshore (13°C) and even cooler offshore (8.5°C) Lake Michigan. Hence, river outflow will flow through the Main Gap sill as a lens over the top of the receiving water (Figure 7 above). These physical conditions allow for harbor water and its constituents to gain access to nearby water treatment plant intakes. In Milwaukee, excursions of conductivity at the Linnwood plant ($+ < 100 \mu\text{Siemens/cm}$ vs. $325 \mu\text{Siemens/cm}$ for open lake water and $700\text{--}1000 \mu\text{Siemens/cm}$ for the harbors) for a day to days are of a magnitude consistent with this model. Seasonality is a crucial component because temperature-density differences between inflows and receiving waters vary tremendously depending on whether they are similar (e.g., mid-summer) and enable mixing, or vastly different (e.g., winter-spring, fall) and encourage strong stratification.

5.1 | Transects Reveal Unseasonable Phytoplankton Enrichment

A relevance of following the flood offshore and over time is that satellites and surface samples sometimes cannot resolve critical features even a few meters underwater, but they can sometimes identify where they are likely to be. In this case, a wave of surface phytoplankton growth seemingly disappeared (e.g., Northeast Reef on July 21, 2008; Figure 8), but in fact only descended to a density shelf in the 20m range where it became a Deep Biomass Maximum (Section 6). We use biomass because sonar scattering at 200 kHz was also dense at the fluorescence maximum, suggesting herbivorous zooplankton or small planktivorous fish such as alewife, and such communities are often found in thin layers moving in response to physical and biological features (Durham and Stocker 2012). Water treatment plant intakes are never at the surface so their depth in midlatitude lakes would be near or just under seasonal thermoclines and

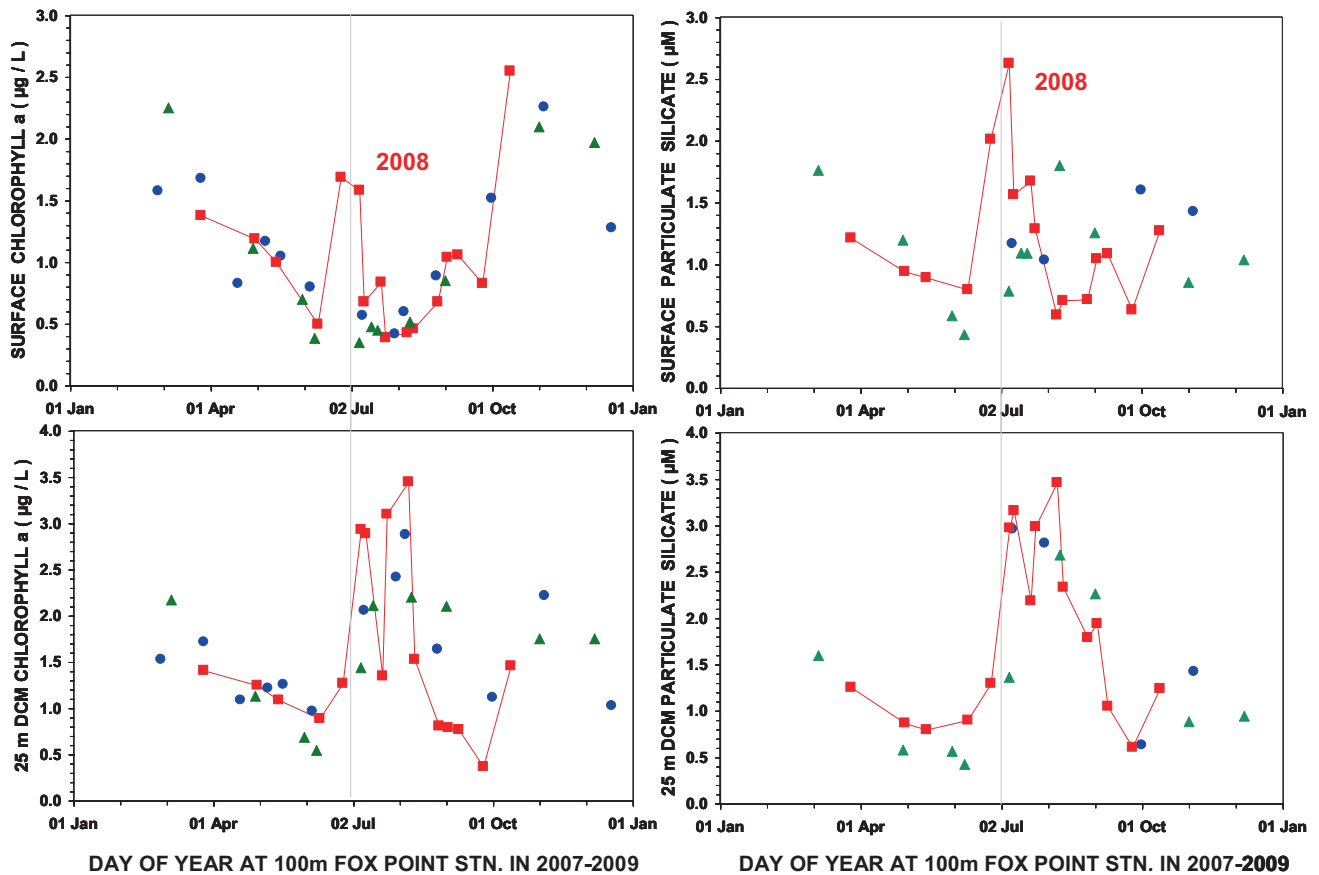


FIGURE 9 | Fox Point 100 m time series station chlorophyll *a* (left panels) and particulate silicate (right panels) concentrations for surface bucket (top panels) and 25 m deep Niskin samples (bottom panels) throughout the years of 2007 (circles, blue), 2008 (squares, red), and 2009 (triangles, green). A light gray line on July 02 aids in visualizing deep sample time lags from sinking blooms during 2008.

therefore also at the depth of subsurface plankton communities and steep chemical gradients.

5.2 | Milwaukee to Northeast Reef and Sheboygan Reef

During the flood year, phytoplankton biomass parameter chlorophyll *a* (all algae) was distinctly enhanced in 2 m surface seawater pump onshore-offshore transects relative to bracketing years 2007 and 2009 (Figure 8). Preceding and following year examples provide timely context for *episodic* change during a simultaneous *chronic* ecosystem disruption from Quagga Mussel establishment. Particulate silica (primarily diatoms) followed a similar pattern for NorthEast Reef, but was weaker at 90 km offshore Sheboygan Reef. The first transect to the NorthEast Reef 40 km offshore was on July 10, a month after the initial flush of runoff, chosen to coincide with a similar transect the previous year, and has since been repeated on that date ± 1 day for 10 years. Chlorophyll *a* and particulate silica were both nearly three times the concentration of bracketing years at the most offshore stations and were higher than bracketing years throughout the transect. The furthest distant points (line from Milwaukee Harbor main gap, Figure 3) at each reef are from stations in the deep basins north of the reef and become shorter moving south (See Figure 3).

The North-of-Peak stations (greatest distance from Milwaukee, Figure 3) have a tendency to be slightly lower in biomass than those on the rise because of current-driven upwelling in the Southerly-flowing current (see Figure 7 Model in Cuhel and Aguilar 2013), then drop again over and behind (South) the reefs because of Quagga Mussel grazing on overflowing upper water (airplane wing effect). The aim of this sampling component, however, was to demonstrate the extent of cohesive transport and the resulting large lake alteration imparted by the flood.

The seeming wave of floodwater moved offshore from the Milwaukee Harbor towards the midlake at a minimum of 1 km per day (39 km in 30 days). The July 10 cruises always worked the Northeast Reef extensively and so characteristics further offshore were not assessed. Both chlorophyll *a* (all phytoplankton) and particulate silica (primarily diatoms, not shown) were highest at the furthest offshore sampling in the two long transects. Of particular note is that during successive offshore transects, previously high surface water phytoplankton (e.g., < 40 km East-North-East) had declined in abundance as further offshore populations grew out. In the following section, the timing of descent is shown to be the causative factor. Similar transects (Figure 8; filled is Northeast Reef, open is Sheboygan Reef with a stop at a Northeast Reef station) accomplished in bracketing years were far lower in each of these parameters.

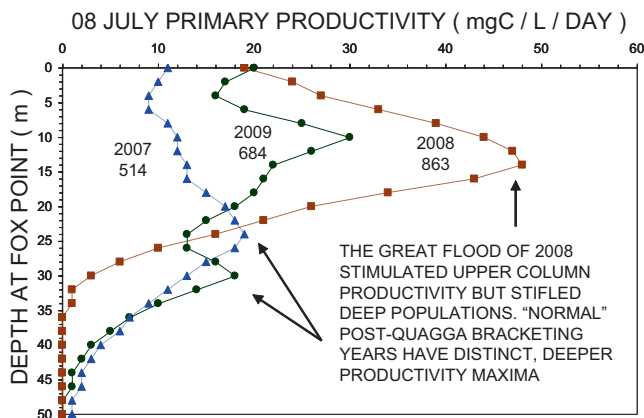


FIGURE 10 | Vertical profiles of modeled CO_2 fixation into organic matter by phytoplankton at the Fox Point time series station for the July 8–10, sampling window in 2007 (circles), 2008 (squares), and 2009 (triangles) based on 8–12 photosynthesis-irradiance sample measurements. Integrated daily areal new organic carbon production in $\text{mgC}/\text{m}^2/\text{day}$ is given under each date.

6 | Fox Point Time Series of Plankton Wave

The wavelike nature of the flood discharge into Lake Michigan, suggested above by the offshore-traveling peak of phytoplankton biomass in transects, was well revealed again in time series profile sampling at the Fox Point reference station, much closer to the water intakes but still upstream of the predominant current. A distinct pulse of chlorophyll *a* and particulate silicate was seen at the surface starting 15 days after peak flow began and lasting less than 2 weeks (Figure 9). There was a coincident increase in Particulate P (PP; plankton biomass here) to concentrations also above seasonal lows in bracketing years (not shown). Both CHL*a* and pSIL tripled between 10 and 25 June, with pSIL lagging 12 days to maximum, while PP increased only 35%, demonstrating substantial plasticity in biomass elemental quotas. That is, the amount of, for example, P in a cell might be more than 100 times the requirement for that cell, which enables that cell to undergo up to seven doublings of growth with no external nutrient availability. This is necessary in a habitat like offshore Lake Michigan where both phosphorus (c.f., Brooks and Edgington 1994; Lean and Cuhel 1989) and some critical trace metal cofactors may be severely limiting yield. To offset this unavailability, phytoplankton may accumulate reserves of these components in rich river-harbor water for continued grow-out. All three biomass parameters decreased suddenly over the 3-day window until the next sampling, returning to levels characteristic of or even lower than bracketing years.

Deep Biomass Maxima (DBM) developed soon after the initial bloom of phytoplankton at the surface (Figure 9 lower panel) for both total algae (chlorophyll *a*) and diatoms (particulate silicate). Development and vertical movement of thin layers of plankton, including both phytoplankton algae and zooplankton consumers, is a common occurrence when density structure exists (McManus et al. 2003; reviewed in Durham and Stocker 2012) and operates over time scales from minutes to months in Lake Michigan. Even allowing for in situ growth,

the magnitude of deep populations sufficiently exceeded surface concentrations to indicate sinking and then productive residency (Figure 10) at the cool water density shelf. This is the most consistent explanation for the apparent disappearance of surface phytoplankton between the 10 and 21 July transect offshore to the reefs as well as the temporal spike during the time series at the Fox Point station. Resulting high areal photosynthetic production (Figure 11 below) lagged behind and persisted twice as long as the surface manifestation, driven by growth of algal communities that sank out of the surface and grew through the summer at depth.

No pulse of elevated dissolved nutrient concentrations was associated with floodwater outflow, though total loading may be higher. It is the concentration that initiates often undesirable algal growth. Examination of Table 2 shows that on the last days of the deluge (June 10, 2008), only LW20, the closest station outside of the harbor, had any noticeable elevation of nutrients that was never observed at the Fox Point reference station. Nitrate, never a limiting nutrient at $>20\mu\text{M}$, decreased consistently within the envelope of bracketing year values until late summer, when it remained lower than 2007 and 2009. Ammonium, a nutrient usually concentrated in river-harbor effluent, was so diluted during flood runoff ($<7\mu\text{M}$) that it could contribute little to the nitrogen budget and rose only $0.6\mu\text{M}$ in the lead-up to surface biomass maxima. It declined rapidly during the maximum and then rose again, but spanning a range of $<1\mu\text{M}$ change. Orthophosphate (SRP) rose 15nM to only 35nM , and the subsequent 30nM decrease was similar to the increase in PP. Both ammonium and phosphate are, however, at the very low concentrations characteristic of recycling economies, and the most important observation is that at 25 km offshore, there is little available. In offshore waters during stratified conditions, concentrations of nutrient compounds nitrate and silicate have no source of replenishment except rain (no Si, some NO_3^-), and hence mixed layer concentration changes integrate consumption over time.

7 | Photosynthesis and Primary Productivity

Usually spring comes a little later in the deeper ($>100\text{m}$) more offshore (25–50 km) waters, but by July deep maxima of various kinds have begun development (Fahnenstiel et al. 2010; Evans et al. 2011). Since Quagga Mussel establishment after 2003, deeper waters below 25–30 m have become the domain of picocyanobacteria (Cuhel and Aguilar 2013) at densities great enough and photoacclimation (ability to use low light) strong enough to undertake decent productivity deep in the water column. Bracketing years 2007 and 2009 showed distinct deep productivity maxima at 32 and 27 m respectively (Figure 10). In 2008, however, diatom communities, as evidenced by, for example, greatly increased particulate silicate, provided enough light absorption to shade out cyanobacterial deep populations. Large diatom biomass in 12–16 m produced a single broad deep photosynthetic maximum also observed during later (July) maturation of further offshore diatom deep maxima. There was a partial recovery of the picocyanobacterial community at 25 m in 2009, when weaker shading by upper water column phytoplankton enabled the deeper maximum to return. Areal production (total carbon fixed under

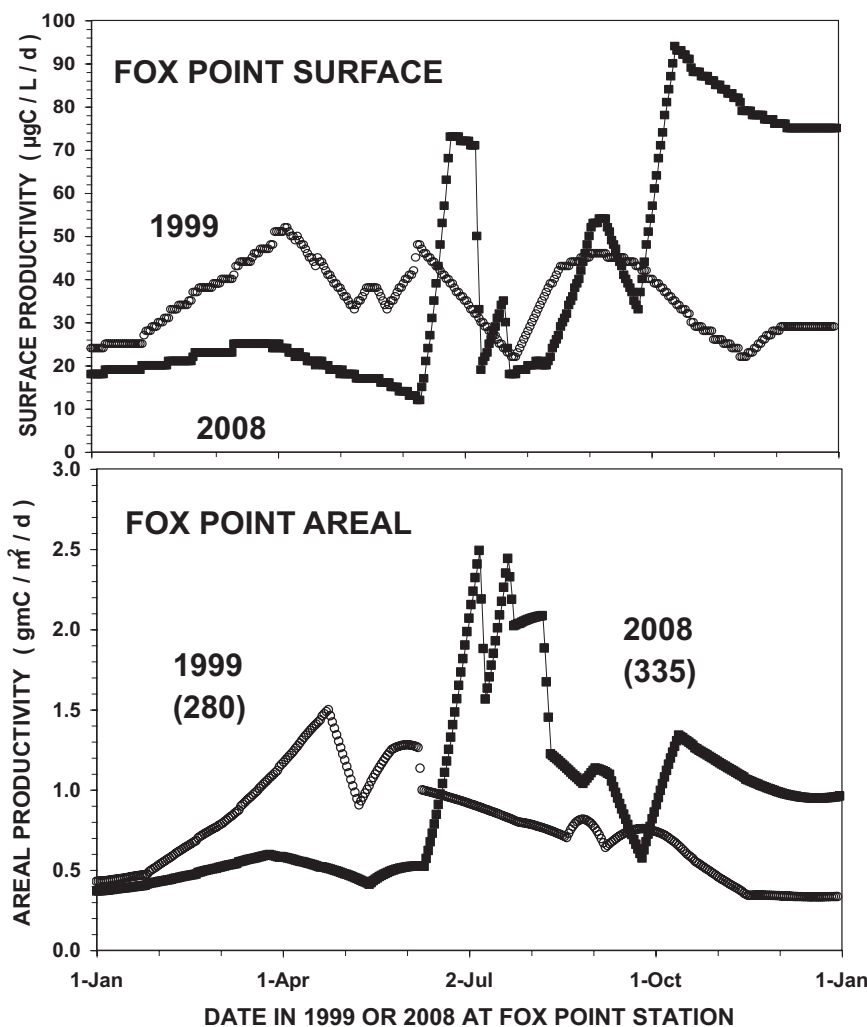


FIGURE 11 | Full year primary productivity at the Fox Point time series station for pre-Quagga Mussel establishment (1999, open circles) and flood year (2008, closed squares) using input data from 18 expeditions (Table 1) throughout the year as exemplified in Figure 8. Top panel, daily surface water volumetric photosynthesis per liter at surface irradiance; bottom panel, areal productivity per square meter to 0.5% surface PAR, with annual carbon fixation in grams C per square meter in parentheses.

one square meter of surface) was 25%–50% greater on this day in 2008 than in bracketing years (under date in Figure 10). More importantly, the greatest part of the production was by large, readily preyed upon diatoms (juicy steaks) in near-surface water instead of picocyanobacteria (sunflower seeds in the shell).

Recently some water producers have noticed an increase in ozone demand (removal of dissolved organic substances and pathogen inactivation) that was NOT associated with either conductivity anomalies or turbidity. Here, a community structure change in which phytoplankton diversity declined in favor of diatom productivity may be at play. Diatoms are encased in a silicate-glass frustule, or outer structure, that makes them dense. In order to maintain themselves in the light they need for photosynthesis, they produce copious quantities of neutral triglycerides, that is, fats, which form buoyant droplets within the cells. When these are eaten by sloppy herbivores, the oils come out and calm surface water (Wang et al. 2025). They can also be released during turbulence in sand filtration and are attacked by ozone to fit observations of recent times.

Absolute productivity is, however, important to the food web, and diatoms can be especially valuable under appropriate conditions (Fee et al. 1992). Altered community structure of phytoplankton species completely changes almost every aspect of ecology (size, light adaptation capability, nutrient use, toxicity, nutritional value, edibility, and others). Since 2006, diatoms have been critically endangered by Quagga Mussel grazing during the mixing season (c.f., Fahnenstiel et al., Fahnenstiel et al. 2010; Kerfoot et al. 2010). Hence, opportunities to escape predation are adaptive, and a warm lens of harbor origin flushing over coastal water provides sunlight, buoyancy, and (stored) nutrients for explosive growth.

Large but seemingly short-lived anomalies in surface biomass were followed by persistent subsurface population enhancement as sinking produced density-shelf Deep Biomass Maxima and especially Deep Particulate Silicate Maxima phytoplankton accumulations (Figure 9). After 5 years of Quagga Mussel-driven phytoplankton depletion (2003–2008, Cuhel and Aguilar 2013), these events created a completely anomalous seasonal and spatial primary productivity regime

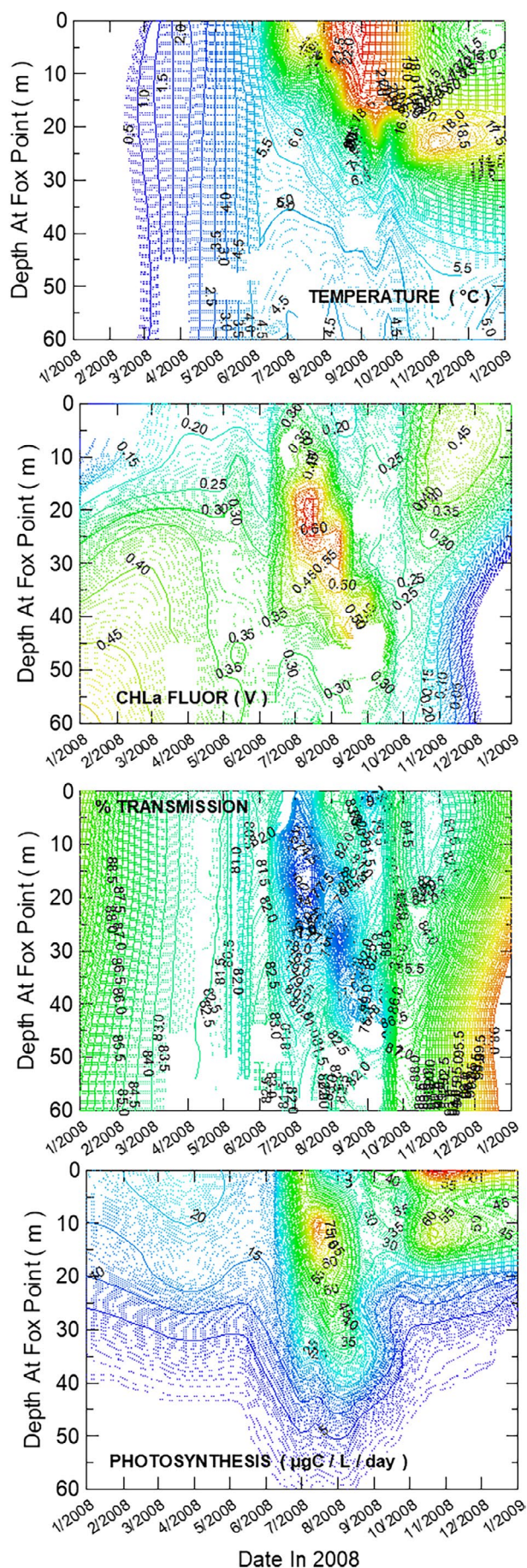


FIGURE 12 | Annual depth contours to 60m for the Fox Point time series station during 2008. Red is hot, blue is not. Temperature panel, vertical lines indicate deep mixing, range 0.5 (blue) to 23°C (red); Chla panel, deep chlorophyll maximum mid-July below 20m, range 0.05 (blue) to 0.6µg Chla/L (red); Transmission panel, blue is turbid and is closely related to chla, range 99.5 (red) to 70.0%T (blue); Photosynthesis panel, major deep growth in July and surface growth in October, range <5 (blue) to 80µg C/L/day (red).

(Figure 11). This figure compares a historically representative pre-Quagga Mussel year (1999) with the Great Midwest Flood of 2008 (5 years into QM establishment) for surface photosynthetic rates and total areal carbon dioxide fixation. Previously, the onset of stratification in May–June resulted in an initial decline of surface productivity as dense diatoms sank out of the upper water column to form a nascent Deep Biomass Maximum. Early warming stimulated a burst of additional productivity that declined to midsummer surface lows. Cooling in the fall reintroduced deep phytoplankton to surface access for a mild bloom that decayed with the waning light of winter.

Both 1999 and 2008 regimes had similar early fall productivity, but 2008 mixing of very shade-adapted deep communities into the upper water column enabled protracted and substantial late fall into early winter biomass growth. Annual production of 335 gC/m²/year in 2008 was 20% higher than the 280 gC/m²/year for the 1999 reference cycle.

Based on 16 visits to the Fox Point time series station in 2008 (Table 1), most with profiling by electronic instrumentation and water collection for biogeochemistry, we have constructed full year contour plots of some crucial parameters to identify their contribution to local aquatic ecology (Figure 12). Blue is low, red is high. Showing only the most active 60m of the 100m water column, we see seasonal changes that are interpretable within the context of described studies above, yet deviate significantly from long-term observations. This program is now (2024) in its 28th year, and we can see change because “change means that you know what was there before” (Cuhel and Aguilar 2016).

The top panel, temperature, shows deep mixing until the first week of June, late warming in late July, and an interesting mid-water temperature inversion in Oct–Nov that could be due to sudden and rapid cooling of fresher surface water. Chlorophyll *a* (Figure 12, second panel) delineates the sudden surging of near-surface phytoplankton in early June followed by rapid descent into midwater. Late summer was characteristically depauperate of phytoplankton in the upper 25–30 m. An unusually strong Fall resurgence placed shade-adapted algae at the surface during the lower light of the season. These primarily diatom algae at the surface are free to circulate widely with wind and currents and are inaccessible to removal by bottom-dwelling filter feeders until deep mixing. Transmission (clarity, third panel) was lowest (blue) in the Deep Biomass Maximum waters (ca. 15–25 m) during a narrow time window in July but distributed to moderate levels throughout the water column in late summer. The late summer–early Fall surface growth is especially evident

in October. Algal photosynthesis (primary productivity) had a strong maximum in the 10–20 m DBM throughout July, and a very strong but thin productivity zone was evident at the surface in October. Hence foraging by small fish and seed populations for spring was unusually robust in 2008.

8 | Conclusions

The Great Midwest Flood of 2008 was a then-extreme meteorological event that led to changes in phytoplankton community structure, lake water optical characteristics, and ultimately ecological progressions in Lake Michigan. Rapid offshore movement (ca. 1 km/day or 0.6 mile/day) of a surficial lens produced an unseasonably strong 2-month near-surface phytoplankton bloom over deep water that was resistant to mussel grazing through stratification-separated surface water and benthos, leading to improved recruitment and growth of planktivorous forage fish. Diatom resurgence occurred at least as far as mid-lake, over 90 km offshore, confirming the potential integrity of water mass transport over long distances.

Milwaukee has one of the world's best designed harbors for nutrient processing from both agricultural and civilization sources, and the Milwaukee Metropolitan Sewerage District (MMSD) is acclaimed as a leader worldwide for reduced release of eutrophicating nutrients and bacteria from sewage treatment. Over the years Milwaukee Harbor effluent, including treated sewage water from MMSD, has been very rarely detectable at only 6 km (3.7 mi) outside of the harbor. Nonetheless, likely river-harbor sources, like the *Cryptosporidium* outbreak of 1993 (Mac Kenzie et al. 1994) that sickened over four hundred thousand people in Milwaukee, require constant vigilance and rapid response capability. The causative agent may have been borne by river effluent moving with the prevailing southerly-flowing current to the Howard Avenue (HA) Water Treatment Plant (Figure 3). Preventative action has responded to post-facto research outcomes to prevent such episodes, including previous construction of the Milwaukee Deep Tunnel, a 400 million gallon underground temporary storage facility for combined sewer overflow waters that enables later processing of street and home-business wastewater through MMSD following inundation events. This greatly reduced incidents of basement backups, sewer overflows and runoff into Lake Michigan, but agricultural runoff was not addressed. Still, such constructs are applicable in many urban venues. Ozone treatment for both pathogens and certain organic materials has also been introduced to reduce or control toxicity. Herein, we have shown, using temperature, conductivity, chemical and biological analysis, and optical profiles, that warm water with river-harbor characteristics can skim over the surface of cooler lakewater to oppose the subsurface currents and distribute solutes and particulates to distant locations.

Regional climate change has become more apparent with each passing year. Currently pressing issues include occasional great floods instead of consistent low-level precipitation. This 2008 example affected biogeochemical and ecological processes far into Lake Michigan, but also in anomalously widespread regions of nearshore water where municipal water intakes are typically found. Its appearance in the physics and chemistry

of the Fox Point reference station 25 km distant from the Main Gap source emphasizes the integrity of water mass composition during long-distance transport. An exacerbating phenomenon is the increased frequency and magnitude of South winds that bring not only unseasonably warm, humid air to the region but also induce western shore coastal upwelling through Ekman Transport (Xie and Hsieh 1995; Gelpi 2023). These upwelling events transport bottom water from mussel beds and harbor effluent into the depth strata of intakes. Both of these operate on shallow nearshore waters where urban water treatment plant intakes are usually found.

The Great Midwest Flood of 2008 produced enough rainwater runoff to create a distinct alteration of Lake Michigan surface water to a distance of 100 km or more. From the Milwaukee River, the 220 million cubic meters of water released in June would make a 10 cm-deep lens 10 km wide that would stretch across the 200 km of lake width. While it is unlikely that this actually occurred, the satellite images and transect sampling unequivocally indicate a basin-scale climate event arising from the sum of all riverine inputs. In addition, a little-considered source, which is direct precipitation onto lake surfaces, was found to be readily detectable as significantly decreased near-surface conductivity at the offshore Fox Point time series station on June 10 prior to the arrival of flood outflow (Figure 5).

Because these processes and others have components at the 10–40 m depth range, understanding their signals is of value for source water quality protection. Besides conductivity anomalies more and more frequently reported by water producers, other subsurface ecological factors may play insidious roles in water quality. Few human populations will continue to have the luxury of uncontaminated potable water supplies as withdrawal and disposal persist. Adequate pre-emptive measurement of water quality proxies provides an opportunity to maintain excellent water provision at a high level from larger water sources like large lakes worldwide.

9 | Recommendations

The above work provides a view of one type of event that can have consequences for users of external freshwater resources. Even deep intakes, as proposed for German Water Producers on Lake Constance (80 m, 2024 workshop in Milwaukee, c.f., Haltiner et al. 2022), must contend with winter, when lakes of any depth circulate top to bottom. It is necessary to either predict or rapidly respond to potential systematic variations.

- This storm sequence demonstrates that during high flows conservative solutes can move 25 km or more while retaining detectable cohesiveness, even after 15 days or more of open water transit. This proves the likely occurrence of nutrient, contaminant, and particle transport from remote sources during flood events. Tracking of Internet output from upstream river gauges on local or nearby sources (USGS, for example, Figure 4) can provide warning of impending flow changes hours to days in advance. Indeed, this event may have incited the USGS to install discharge gauge USGS-04087142 on the Menomonee River, a tributary to the Milwaukee Harbor, which began reporting on October 06, 2008, shortly after peak flows of this epic flood.

- Temperature and conductivity are crucial factors in understanding freshwater sources, water movement, or stability. Seasonal progressions of temperature at intake depths are often punctuated by anomalies induced by upwelling and storm mixing, for example. Upwelling can protect intakes, usually located on the bottom of large lakes, because off-shore cold bottom water does not contain terrestrially-originating chemical and biological threats common to sudden voluminous runoff. Hence, alerts of suddenly decreasing intake temperatures can identify periods of upwelling that may protect intake stability. Less appreciated but of substantial value is knowledge of water column structure and physics–chemistry of nearby *potential* source waters. Layering of river and harbor waters, usually proximal to water producer intakes, controls what materials might move to them. River–harbor and receiving waters are often of very different composition and risk. Their existence may be revealed by differences of temperature and/or conductivity readily monitored by sensor strings. A temperature and/or conductivity meter deployment with cabled readout can be established with one sensor laying on nearby sediment and one sensor at near-intake depth (attached to intake or sub-surface buoy). The latter already exists in many installations. Systematic differences between them can be ascribed to different water origins and hence compositions. Interpretation can be coordinated with collaboration with local sewage districts that must regularly survey water quality in the vicinity of their outfalls.
- Planning response to impending flood events should consider seasonality initially in mid- and high-latitude locations. In Spring, runoff during storms is warmed by ground contact while large receiving waters remain cold. As a result, outflows will float and entrainment risk is reduced. In contrast, late Fall storms meet cooler surfaces while lake-waters cool slowly, enabling river-harbor waters to flow along bottom and at the level of intakes.
- Preceding and following year examples provide timely context for *episodic* change during simultaneous *chronic* ecosystem disruptions, here from Quagga Mussel establishment. The same may be true for initiation of major direct human interference, such as Milwaukee's impending 5-year-long dredging of over 700,000 m³ (1 million cubic yards) of river and harbor sediments and deposition in retention ponds upstream of harbor outflows. Knowing in advance that soluble eutrophicating nutrients, metals, and dissolved gases will be chronically elevated for half a decade can assist in preparation. Open water nutrients and plankton communities progress on days to weeks scales and predictions can be made using accessible past records of weather-driven events. Investigate cause-and-effect relationships among water chemistry, meteorology, and biological features for past events and consider them ahead of impending change.
- No pulse of elevated dissolved macronutrient *concentrations* (e.g., N, P, S, Si) to receiving water was associated with floodwater outflow, as is common for high runoff events. Alternatively, very limiting micronutrients such as reduced iron, manganese, and selenium that stimulate phytoplankton growth may be released from turbulently resuspended legacy sediments in the flow field. Often flow concentrations are lower during floods from dilution greater than weathering, and though mass flux may be much higher it is the concentration that affects micro-biology and stormwater management the most.
- Awareness of loading capacity of decontamination systems (e.g., charcoal, ozone, filtration) and projection of impending loading from records of experience can help prepare for events. Case in point: COVID-era lack of access to bleach created difficulty with mussel veliger (swimming larval stage) eradication and required judicious use of limiting resources.
- Transmission (%*T*) and reflectance turbidity (NTU, Nephelometric Turbidity Units) measure different particle characteristics and together can discern suspended clays and carbonates from biological suspensions. %*T* is always much more sensitive to particles but also responds to colored dissolved organic matter. Turbidity sensors respond to *reflected* light, common for white materials like carbonates and clays but insensitive to absorptive materials like algae, while transmission is less reduced by white particles that can reflect *forward* but is sensitive to absorbing particles like algae and dirt. The ratio of %*T*/NTU is roughly diagnostic for suspended material types. Flow-through transmissometers as found on our CTD instruments are readily available for intake type systems and require little maintenance with a high-quality wiper or regular service. NTU sensors are already in action in most water intake systems.
- Particulate silicate is an excellent measure of developing diatom populations to warn against impending filtration and UV oxidation system stresses. It is nearly as simple as dissolved silicate and is measured, like phosphate, by chemical reaction amenable to autoanalyzer analysis. It requires two additional steps (filtration, extraction) and can be accomplished daily by typically trained bench chemists. Because of its high sensitivity, early awareness of plankton community progressions can be obtained.
- Size distribution of particulate material, particularly chlorophyll *a*, is a simple and informative analysis using 10 μm pore size filters versus 0.2 μm or glass fiber filters for either fluorometric or dry weight methods. Typically, low nutrient Oligotrophic ecosystems favor smaller-celled phytoplankton, while rich waters may produce fast-growing, larger celled communities.

Author Contributions

Russell L. Cuhel: conceptualization, data curation, formal analysis, supervision, funding acquisition, investigation, visualization, methodology, writing – original draft, project administration, writing – review and editing. **Carmen Aguilar:** conceptualization, data curation, formal analysis, supervision, funding acquisition, validation, investigation, visualization, methodology, project administration, writing – review and editing.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available on request from the corresponding author. Portions are housed in University databases (waterbase.uwm.edu) which are publicly inaccessible at this time. The authors are working to deposit all data types in the national data repositories operated presently by the National Oceanic and Atmospheric Administration.

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